A safe and effective formalin inactivated, **alum conjugaged vaccine** containing HAV grown in human diploid cell culture. Course consists of **two intra muscular injections** of the vaccine. Protection begins **4 weeks** after injection and lasts for **10 to 20 years**. No specific antiviral drug is available.

Type B Hepatitis (HBV)

HBV is a 42nm **DNA virus** with an outer envelope and an inner core 27nm in diameter. Enclosing the **viral genome** and a **DNA polymerase**. It belongs to the family Hepadna Viridae HBV is **'Hepadna Virus type 1'**. **Australia antigen** was found to be associated with serum hepatitis. It was the surface component of HBV, so named as **hepatitis B surface antigen (HBsAg)**.

3 types of particles are visualized, most abundant form is a **spherical particle**, 22nm in diameter. The second type of particle is **filamentous** or **tabular** with a diameter of 22nm both are antigenically identical. Third type of particle for fewer in number, is a **double walled spherical** structure 42 nm in diameter. This particle is the complete **hepatitis B virus**, known as **Dane particle**.

The envelope proteins expressed on the surface contains **hepatitis B surface antigen (HBsAg)**. HBsAg consists of two major polypeptides, one of which is glycosylated. The nucleocapsid or core contains **hepatitis B core antigen (HBcAg A)** (Figure 10.5)**.** Third antigen called the **hepatitis B e antigen (HBeAg)** is a soluble non particulate nucleocapsid protein.

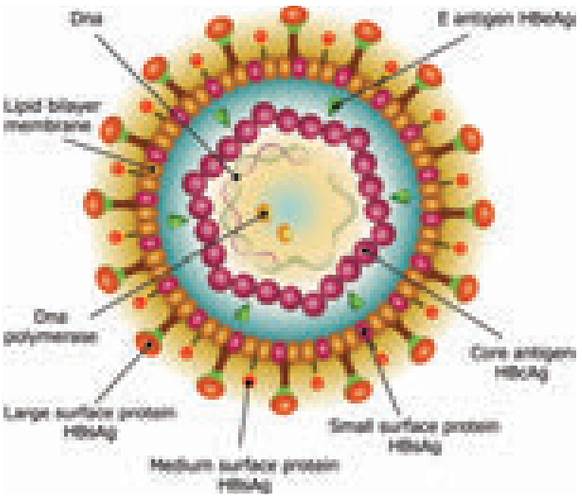


Figure 10.5: Structure of Hepatitis B Virus

The **nucleocapsid** encloses the viral genome consisting of **two linear strands of DNA** held in a circular configuration. One of the strands **incomplete** (+ strand) DNA appears partially **double stranded** and partially **single stranded**. Associated with the + strand is a **viral DNA polymerase** (has both DNA dependent DNA polymerase and RNA dependent reverse transcriptase functions). This polymerase can **repair the gap** in the plus strand and render the **genome fully double straned**.

Natural infection occurs **only in humans**. The virus is maintained in carriers whose blood contains **circulating virus** for long periods carriers are of two categories, the highly infectious **super carriers** and the simple carriers. Former have high titre HBsAg along with HBsAg, DNA polymerase and HBV in ciruculation. Simple carriers have low infectivity and low titre HBsAg in blood.

HBV is a **blood borne virus** and the infection is transmitted by **parenteral, sexual and perinatal models**. The virus may also be present in other **body fluids and excreations** such as saliva, breast

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